

MICROPROCESSORS

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- 1 A microprocessor is a general purpose central processing unit on a single chip that requires external components to function as a computer while a microcontroller is a "computer on a chip" that integrates the microprocessor memory and I/O peripherals onto a single integrated circuit
- 2 Address lines : $2^8 = 256$ unique addresses per chip
Total addresses = $256 \times 4 = 1024$ locations
Word size : 16 bits = 2 bytes
Maximum memory size : $1024 \times 2 = 2048$ bytes
- 3 Accumulator, ALU, General purpose registers, Program counter.
- 4 The 8085 uses the parity flag by setting the parity flag to 1 if the result contains even number of 1s.
- 5 INTA - is a general purpose interrupt signal used to initiate an interrupt service routine.
SID - used for serial data communication
ALE - a positive pulse generated during the first T-state of a machine cycle

6 Clock frequency - 2MHz

Loop count : 2384H = 9092 in decimal

T-states : 6(DCX) + 4(MOV) + 4(ORA) + 10(JNZ) = 24 T-states

Total T-states : 10(LXI) + (9092 x 24) + (6 + 4 + 4 + 10) = 218215 T-states

Total time : 218215 x 0.5 = 0.1091075 seconds

7 • The MPU places the 8-bit port address on both upper and lower address buses

• Sets the $\overline{IO/\overline{M}}$ signal to high indicating an I/O operation

• Sends a pulse on the ALE pin to the latch address.

• Places the data from the accumulator onto the data bus.

• Activates the $\overline{WR}$ control signal, which enables the external output port to latch the data from the bus.

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D7	D6	D5	D4	D3	D2	D1	D0
SOD	SOE	X	R7.5	MSE	M7.5	M6.5	M5.5